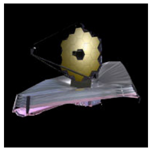

6915 - Direct Detection and Characterization of a Nearby Temperate Giant Planet

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI F1065C				
	1	HD 222 BKG	MIRI Coronagraphic Imaging	(2) HD-222237-BKG
	4	HD 222 FQPM	MIRI Coronagraphic Imaging	(1) HD-222237
	7	PSF-REF FQPM	MIRI Coronagraphic Imaging	(5) HD-22886
	10	PSF-REF BKG	MIRI Coronagraphic Imaging	(6) HD-22886-BKG
MIRI F1140C				
	2	HD 222 BKG	MIRI Coronagraphic Imaging	(2) HD-222237-BKG
	5	HD 222 FQPM	MIRI Coronagraphic Imaging	(1) HD-222237

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
	8	PSF-REF FQPM	MIRI Coronagraphic Imaging	(3) HD-218288
	11	PSF-REF BKG	MIRI Coronagraphic Imaging	(4) HD-218288-BKG
MIRI F1550C				
	3	HD 222 BKG	MIRI Coronagraphic Imaging	(2) HD-222237-BKG
	6	HD 222 FQPM	MIRI Coronagraphic Imaging	(1) HD-222237
	9	PSF-REF FQPM	MIRI Coronagraphic Imaging	(5) HD-22886
	12	PSF-REF BKG	MIRI Coronagraphic Imaging	(6) HD-22886-BKG
NIRCam F444W				
	13		NIRCam Coronagraphic Imaging	(1) HD-222237
	14		NIRCam Coronagraphic Imaging	(1) HD-222237
	15		NIRCam Coronagraphic Imaging	(3) HD-218288

ABSTRACT

We propose to image HD 222237 b, a nearby, temperate, eccentric giant planet prime for characterization by JWST. HD 222237 b, observable only with JWST, is the next rung on the "effective temperature ladder" that will connect known directly imaged planets (like eps Indi b) and Y-type brown dwarfs (like W0359 or W0855) to Jupiter and Saturn. It provides a rare opportunity to study the atmospheric state of giant planets in the asymptotic phase of their evolution and benchmark evolutionary models, and will be a critical point of comparison between directly imaged planets and mature, temperate transiting exoplanets. This program will target the potentially coldest-imaged planet with the NIRCam and MIRI coronagraphs. These observations will constrain key atmospheric model uncertainties, like the strength of water-ice cloud opacity, the abundance of ammonia, and the strength of disequilibrium chemistry in the planet's atmosphere. This program is designed to efficiently detect the planet at high confidence, photometrically characterize the atmosphere, and refine the planet's sky-projected orbit ahead of Cycle 5; doing so will allow the community to estimate the feasibility of follow-up spectroscopy on the fastest timescale.

OBSERVING DESCRIPTION

In order to constrain 1) the cloud strength, 2) the ammonia abundance, and 3) the dis-equilibrium chemistry of an exoplanet with Teff <250 K for the first time, we will observe HD 222237 b with JWST /NIRCam and MIRI coronagraphy. These observations will each require corresponding PSF reference star observations that will use the 9-point dither patterns to increase the diversity of the PSF library used for starlight subtraction with principle com-ponent analysis. Due to the presence of stray light “glowsticks” in the MIRI FQPMs, the

MIRI observations will require the observation of dedicated background observations.

We optimized exposure patterns and times using pancake and pynrc for MIRI and NIRCam respectively to achieve 5-10% photometry in the MIRI filters, and a 3-5 sigma detection in the NIRCAM LW filter.

We selected HD 218288 as a reference star for both instruments, it is nearby, brighter than HD 222237 b by 1.5 mags, and of a similar spectral type (K3III vs K3V).

Based on the preliminary orbit fit, the MIRI FQPM observations have the aperture PA restriction 281-301 deg, to ensure the planet does not fall behind the edges of the coronagraph regardless of the degenerate PA angle (either 63 or 153 deg). Once candidate sources close in are found, this restriction can be revised to increase schedulability.

We will execute the F1140C observations (where the planet should be secondmost bright) first in order to determine the positions of candiate sources and whether our orbital predictions are correct. If we detect the planet in F1140C, we expect to be able to see the planet easily in both F1065C and F1550C; if we cannot detect the planet at >5 sigma in F1140C, we will allocate the F1065C time to the F1550C filter, in anticipation of only recovering the planet in the longest filter. This failstate is not favored by our modeling, but is conceivable.

Then, the F1065C and F1550C observations will be executed during the target's next observability window, to observe common proper motion (~0.5-1 MIRI pixels) and measure the MIR color of any candidates. This minimizes the total number of telescope slews while preserving our multi epoch observing strategy. The NIRCam observations, without aperture PA requirements (only relative PA restrictions for ADI) will be allowed to float to increase schedulability.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems]				
(2)	HD-222237-BKG	RA: 23 39 16.4440 (354.8185167d) Dec: -72 42 28.13 (-72.70781d) Equinox: J2000	Epoch of Position: 2000	
Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4) Category=Unidentified Description=[Blank field]				
(3)	HD-218288	RA: 23 08 35.5958 (347.1483158d) Dec: -73 35 10.03 (-73.58612d) Equinox: J2000	Proper Motion RA: 22.994 mas/yr Proper Motion Dec: -2.554000070631446 mas/yr Parallax: 0.004196500000000001" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K giants, K stars]				
(4)	HD-218288-BKG	RA: 23 08 54.8830 (347.2286792d) Dec: -73 33 40.11 (-73.56114d) Equinox: J2000	Epoch of Position: 2000	
Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4) Category=Unidentified Description=[Blank field]				
(5)	HD-22886	RA: 03 41 18.0743 (55.3253096d) Dec: +25 00 29.39 (25.00816d) Equinox: J2000	Proper Motion RA: 13.029 mas/yr Proper Motion Dec: 18.363 mas/yr Parallax: 0.0050692" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars]				
(6)	HD-22886-BKG	RA: 03 41 15.6600 (55.3152500d) Dec: +24 58 3.30 (24.96758d) Equinox: J2000	Epoch of Position: 2000	
Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4) 03 41 15.66 +24 58 03.3 Category=Unidentified Description=[Blank field]				

Observation

Proposal 6915, Observation 1: HD 222 BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [HD 222 FQPM (Obs 4)]

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-222237-BKG	RA: 23 39 16.4440 (354.8185167d) Dec: -72 42 28.13 (-72.70781d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1065C	MASK1065	4QPM	F1065C	FASTR1	1251	29	1	1	29	8702.062	

PSF References

Additional Justification: false

Special Requirements	
	No Parallel Attachments Sequence Observations 1, 4, 7, 10, Non-interruptible

Observation

Proposal 6915, Observation 4: HD 222 FQPM

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[HD 222 BKG (Obs 1)]

Diagnostics

(HD 222 FQPM (Obs 4)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	22	1	1	5.273	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1065C	MASK1065	4QPM	F1065C	FASTR1	1251	29	1	1	29	8702.062	

PSF References	PSF-REF FQPM (Obs 7) (PSF Reference; Filters [F1065C]) Additional Justification: false
Special Requirements	Aperture PA Range 0 to 25 Degrees (V3 355.16455103 to 20.16455103) No Parallel Attachments 4 After 5 by 180 Days to <None specified> Sequence Observations 1, 4, 7, 10, Non-interruptible

Observation

Proposal 6915, Observation 7: PSF-REF FQPM

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[PSF-REF BKG (Obs 10)]

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	HD-22886	RA: 03 41 18.0743 (55.3253096d) Dec: +25 00 29.39 (25.00816d) Equinox: J2000	Proper Motion RA: 13.029 mas/yr Proper Motion Dec: 18.363 mas/yr Parallax: 0.0050692" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K stars]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	8	1	1	1.917	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	9-POINT-SMALL-GRID

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1065C	MASK1065	4QPM	F1065C	FASTR1	1228	5	1	9	45	13253.345	

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 1, 4, 7, 10, Non-interruptible

Observation

Proposal 6915, Observation 10: PSF-REF BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [PSF-REF FQPM (Obs 7)]

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	HD-22886-BKG	RA: 03 41 15.6600 (55.3152500d) Dec: +24 58 3.30 (24.96758d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)

03 41 15.66 +24 58 03.3
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1065C	MASK1065	4QPM	F1065C	FASTR1	1228	5	1	1	5	1472.594	

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 1, 4, 7, 10, Non-interruptible

Observation

Proposal 6915, Observation 2: HD 222 BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [HD 222 FQPM (Obs 5)]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-222237-BKG	RA: 23 39 16.4440 (354.8185167d) Dec: -72 42 28.13 (-72.70781d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	27	1	1	27	8101.903	

PSF References

Additional Justification: false

Special Requirements	
	No Parallel Attachments Sequence Observations 2, 5, 8, 11, Non-interruptible

Observation

Proposal 6915, Observation 5: HD 222 FQPM Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[HD 222 BKG (Obs 2)]

Diagnostics

(HD 222 FQPM (Obs 5)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	22	1	1	5.273	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	27	1	1	27	8101.903	

PSF References	PSF-REF FQPM (Obs 8) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	Aperture PA Range 293 to 305 Degrees (V3 288.16455103 to 300.16455103) No Parallel Attachments 4 After 5 by 180 Days to <None specified> Sequence Observations 2, 5, 8, 11, Non-interruptible

Observation

Proposal 6915, Observation 8: PSF-REF FQPM

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[PSF-REF BKG (Obs 11)]

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	HD-218288	RA: 23 08 35.5958 (347.1483158d) Dec: -73 35 10.03 (-73.58612d) Equinox: J2000	Proper Motion RA: 22.994 mas/yr Proper Motion Dec: -2.554000070631446 mas/yr Parallax: 0.004196500000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K giants, K stars]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	8	1	1	1.917	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	9-POINT-SMALL-GRID

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	5	1	9	45	13501.414	

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 2, 5, 8, 11, Non-interruptible

Observation

Proposal 6915, Observation 11: PSF-REF BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [PSF-REF FQPM (Obs 8)]

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	HD-218288-BKG	RA: 23 08 54.8830 (347.2286792d) Dec: -73 33 40.11 (-73.56114d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	5	1	1	5	1500.157	

PSF References

Additional Justification: false

Special Requirements	
	No Parallel Attachments Sequence Observations 2, 5, 8, 11, Non-interruptible

Observation

Proposal 6915, Observation 3: HD 222 BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [HD 222 FQPM (Obs 6)]

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-222237-BKG	RA: 23 39 16.4440 (354.8185167d) Dec: -72 42 28.13 (-72.70781d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1550C	MASK1550	4QPM	F1550C	FASTR1	1251	20	1	1	20	6001.348	

PSF References

Additional Justification: false

Special Requirements	No Parallel Attachments Sequence Observations 3, 6, 9, 12, Non-interruptible

Observation

Proposal 6915, Observation 6: HD 222 FQPM

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[HD 222 BKG (Obs 3)]

Diagnostics

(HD 222 FQPM (Obs 6)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	22	1	1	5.273	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1550C	MASK1550	4QPM	F1550C	FASTR1	1251	20	1	1	20	6001.348	

PSF References	PSF-REF FQPM (Obs 9) (PSF Reference; Filters [F1550C]) Additional Justification: false
Special Requirements	Aperture PA Range 0 to 25 Degrees (V3 355.16455103 to 20.16455103) No Parallel Attachments Sequence Observations 3, 6, 9, 12, Non-interruptible

Observation

Proposal 6915, Observation 9: PSF-REF FQPM

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observations:[PSF-REF BKG (Obs 12)]

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	HD-22886	RA: 03 41 18.0743 (55.3253096d) Dec: +25 00 29.39 (25.00816d) Equinox: J2000	Proper Motion RA: 13.029 mas/yr Proper Motion Dec: 18.363 mas/yr Parallax: 0.0050692" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K stars]

Acquisition

#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	FND	1	FAST	8	1	1	1.917	223489

Template

Repeat observation
NO

Dithers

#	Dither Type
1	9-POINT-SMALL-GRID

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1550C	MASK1550	4QPM	F1550C	FASTR1	1251	5	1	9	45	13501.414	

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 3, 6, 9, 12, Non-interruptible

Observation

Proposal 6915, Observation 12: PSF-REF BKG

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Coronagraphic Imaging

Background Observation For: [PSF-REF FQPM (Obs 9)]

Diagnostics

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	HD-22886-BKG	RA: 03 41 15.6600 (55.3152500d) Dec: +24 58 3.30 (24.96758d) Equinox: J2000	Epoch of Position: 2000	

Comments: Validated as viable background with vizier (allWISE images) and IRSA wise viewer tools, checking longer wavelength channels (ch2-ch4)

03 41 15.66 +24 58 03.3
Category=Unidentified
Description=[Blank field]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Repeat observation	Background Quadrant
FND	NO	1

Dithers

#	Dither Type
1	NONE

Spectral Elements

#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	4QPM/F1550C	MASK1550	4QPM	F1550C	FASTR1	1251	5	1	1	5	1500.157	

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 3, 6, 9, 12, Non-interruptible

Observation

Proposal 6915, Observation 13

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Background Observations:[]

Diagnostics

(Observation 13) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 13) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	SHALLOW2	33	1	1	8.179	223489

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK335R	true	SUB320A335R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	4	1	1	42.947	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	SHALLOW4	10	120	1	120	6416.698	

PSF References	Observation 15 (PSF Reference; Filters [F200W/F444W]) Observation 14 (Filters [F200W/F444W]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 13, 14, 15, Non-interruptible Aperture PA Offset 14 from 13 by 7 to 14 Degrees (Same offsets in V3)

Observation

Proposal 6915, Observation 14

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Background Observations:[]

Diagnostics

(Observation 14) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 14) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-222237	RA: 23 39 37.3874 (354.9057808d) Dec: -72 43 19.76 (-72.72216d) Equinox: J2000	Proper Motion RA: 143.736 mas/yr Proper Motion Dec: -736.9070000095235 mas/yr Parallax: 0.0873724" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Exoplanet Systems]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	SHALLOW2	33	1	1	8.179	223489

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK335R	true	SUB320A335R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	4	1	1	42.947	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	SHALLOW4	10	120	1	120	6416.698	

PSF References	Observation 15 (PSF Reference; Filters [F200W/F444W]) Observation 13 (Filters [F200W/F444W]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 13, 14, 15, Non-interruptible Aperture PA Offset 14 from 13 by 7 to 14 Degrees (Same offsets in V3)

Observation

Proposal 6915, Observation 15

Wed Aug 06 19:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Background Observations:[]

Diagnostics

(Observation 15) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	HD-218288	RA: 23 08 35.5958 (347.1483158d) Dec: -73 35 10.03 (-73.58612d) Equinox: J2000	Proper Motion RA: 22.994 mas/yr Proper Motion Dec: -2.554000070631446 mas/yr Parallax: 0.004196500000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[K giants, K stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	SHALLOW2	17	1	1	4.166	223489

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK335R	false	SUB320A335R	9-POINT-CIRCLE

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	SHALLOW4	6	36	9	324	10397.704	

PSF References

PSF Reference: true

Proposal 6915 - Observation 15 - Direct Detection and Characterization of a Nearby Temperate Giant Planet

Special Requirements	
	Sequence Observations 13, 14, 15, Non-interruptible